

Assignment Kit for Size Counting Standard



Personal Software Process for Engineers: Part I

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Personal Software Process for Engineers: Part I

Assignment Kit for the Size Counting Standard

Overview

Overview This assignment kit covers the following topics.

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Prerequisites Prerequisite reading

- Chapter 3

Objectives The objectives of Size Counting standard are to

- define the size counting standards that are appropriate for the programming language and environment that you will use
- provide a basis for developing a coding standard
- prepare for developing a program to count program size

Size counting standard requirements

Size counting standard requirements

Produce and document a standard for counting program size for the language and environment that you will use in this course.

Submit the completed standard using the format in the template on page 11 with your program 2 assignment package.

Example 1: Pascal size counting standard

Overview

The template is a simplified version of the SEI measurement framework.

- Use this template to describe important items.
- Tailor it to fit your needs or language.

We'll walk through two example size counting standards. The first example is for logical LOC for Pascal programs.

Completing the header

Complete the header as follows:

- the name you give this standard
 - the language you are using
 - your name
 - the date you produced this standard (or revision)
-

Count type

Choices are logical and physical LOC.

- Logical LOC counts language elements.
- Physical LOC counts text lines.

For this counting standard, you are counting logical LOC.

Statement type

Use this section to define how you will count various types of statements.

Consider the following:

- How are you going to count procedure declarations and function prototypes?
 - How will you count compiler directives?
 - Will you count blank lines or comments? Why or why not?
-

Clarifications

A fully operational standard generally requires many notes and comments. Use the clarification section for this purpose.

Continued on next page

Pascal LOC Counting Standard Template

Definition Name:	<i>Example Pascal LOC Std.</i>	Language:	<i>Pascal</i>
Author:	<i>W. S. Humphrey</i>	Date:	<i>12/20/93</i>

Count Type	Type	Comments
Physical/Logical	<i>Logical</i>	
Statement Type	Included	Comments
Executable	<i>yes</i>	
Nonexecutable:		
Declarations	<i>yes</i>	
Compiler Directives	<i>yes</i>	
Comments	<i>no</i>	
Blank lines	<i>no</i>	
Other elements		
Clarifications		Examples/Cases
<i>Nulls</i>	<i>yes</i>	<i>continues, no-ops, ...</i>
<i>Empty statements</i>	<i>yes</i>	<i>“;;”, lone ;’s, etc.</i>
<i>Generic instantiators</i>		
<i>Begin...end</i>	<i>note 1</i>	<i>when executable</i>
<i>Begin...end</i>	<i>note 1</i>	<i>when not executable</i>
<i>Test conditions</i>	<i>yes</i>	
<i>Expression evaluation</i>	<i>yes</i>	<i>when used as sub program arguments</i>
<i>End symbols</i>	<i>notes 1,2</i>	<i>when terminating executable statements</i>
<i>End symbols</i>	<i>notes 1,2</i>	<i>when terminating declarations or bodies</i>
<i>Then, else, otherwise</i>	<i>note 1</i>	
<i>Elseif</i>	<i>note 1</i>	
<i>Keywords</i>	<i>notes 1,2</i>	<i>procedure division, interface, implementation</i>
<i>Labels</i>	<i>yes</i>	<i>branch destinations when on separate lines</i>
Note 1		<i>unless followed by ; or. or included in {}, count the following keywords once: BEGIN, CASE, DO, ELSE, END, IF, RECORD, REPEAT, THEN, UNTIL</i>
Note 2		<i>count every ; and . that is not within a {} or ()</i>
Note 3		<i>count each , between USES and the next ; or between VAR and the next ;</i>

Example 2: C++ LOC counting standard

**How many
LOC?**

Using the C++ LOC counting standard on page 7, how many LOC are in the following program fragment?

```
#include <stdio.h>

void main (void)
{
    int i,j;

    for (i = 0; i < 10; i++)
        for (j = 0; j < 10; j++)
            cout << i << ", " << j << endl;
}
```

Continued on next page

Example C++ Size Counting Standard

Definition Name: *Example C++ LOC std.*

Language: *C++*

Author: *W.S. Humphrey*

Date: *12/20/93*

Count Type	Type	Comments
Physical/Logical	<i>Logical</i>	
Statement Type	Included	Comments
Executable	<i>Yes</i>	
Nonexecutable:		
Declarations	<i>Yes, Notes 3, 4</i>	
Compiler Directives	<i>Yes, Note 4</i>	
Comments	<i>No</i>	
Blank lines	<i>No</i>	
Other elements		
Clarifications		Examples/Cases
<i>Empty statements</i>	<i>yes</i>	<i>";;", lone ;'s, etc.</i>
<i>Begin...end</i>	<i>note 1</i>	
<i>Expression evaluation</i>	<i>yes</i>	<i>when used as sub program arguments</i>
<i>End symbols</i>	<i>notes 1,2</i>	<i>for terminating executable statements, declarations, bodies</i>
<i>Then, else, otherwise</i>	<i>note 1</i>	
<i>Elseif</i>	<i>yes</i>	
<i>Keywords</i>	<i>yes</i>	<i>procedure division, interface, implementation</i>
<i>Labels</i>	<i>yes</i>	<i>branch destinations when on separate lines</i>
<i>Note 1</i>		<i>Count once every occurrence of the following key words: CASE, DO, ELSE, ENUM, FOR, IF, PRIVATE, PUBLIC, STRUCT, SWITCH, UNION, WHILE</i>
<i>Note 2</i>		<i>count once every occurrence of the following: ; , {} or {};</i>
<i>Note 3</i>		<i>count each variable or parameter declaration</i>
<i>Note 4</i>		<i>count once each #define, #ifdef, #include, etc. statement</i>

Example 3: another size counting standard

Overview

For this example standard, the class will select from among the following product categories.

- documents
- interface screens and forms
- database program elements
- maintenance fixes
- any other requested category

We'll then walk through developing the selected size counting standard.

Completing the standard

We'll then walk through the completion of the standard as a class exercise.

Continued on next page

Size Counting Standard Template

Definition Name: Distribucion t Language: java
 Author: Bridget Mendez Date: 2-12-25

Count Type	Type	Comments
Physical/Logical	Logical	El proyecto usa estructuras largas, métodos con lógica matemática y recursión. Contar LOC físicos no representaría adecuadamente la complejidad.
Statement Type	Included	Comments
Executable	si	Incluye return, asignaciones, llamadas a métodos, System.out.println, cálculos matemáticos, recursión (GammaFunction), bucles (for), etc.
Nonexecutable:	si	Declaración de métodos, clases, atributos privados. Necesario porque tu proyecto usa orientación a objetos fuertemente.
Declarations	si	Cada atributo de clase: int intNumSeg;, double db1E;, etc. Cada uno cuenta como 1 LOC.
Compiler Directives	no	Java no usa directivas tipo #include.
Comments	no	Tus archivos contienen encabezados de copyright, avisos legales y documentación. Todo eso NO se cuenta.
Blank lines	no	Aparecen en todos tus archivos para separar secciones. No cuentan.
Other elements	Sí, pero seleccionados	Reglas específicas en Clarifications.
Clarifications		Examples/Cases
1. Declaraciones de clase		archivos cada clase inicia así: • public class App { • public class Logic { • public class SimpsonIntegration {
2. Atributos privados		private int intNumSeg; private double db1E; private int intDOF;
3. Constructores		public Logic() { this.intNumSeg = 10; this.db1E = 1E-5; }

Note		
Note		
Note		
Note		

Evaluation criteria and suggestions

Evaluation criteria

Your standard must be

- complete
 - legible
-

Suggestions

Keep your standards simple and short.

Do not hesitate to copy or build on the PSP materials.

Size Counting Standard Template

Definition Name: _____ Language: _____
 Author: _____ Date: _____

Count Type	Type	Comments
Physical/Logical		
Statement Type	Included	Comments
Executable		
Nonexecutable:		
Declarations		
Compiler Directives		
Comments		
Blank lines		
Other elements		
Clarifications		Examples/Cases
Note		
Note		
Note		
Note		